

UMM Unlearning

Knows But Can't Draw: Toward Modality-Selective Concept Suppression in Unified Multimodal Models

May 13th, 2026

Hayeon Jeong

Base Model Architecture

- **UMM (Unified Multimodal Models)**

- One model handles text & image tokens.
- Text tokens support understanding.
- VAE tokens drive image generation.
- The tokens meet in a shared transformer.
- We hook the VAE-side hidden state.

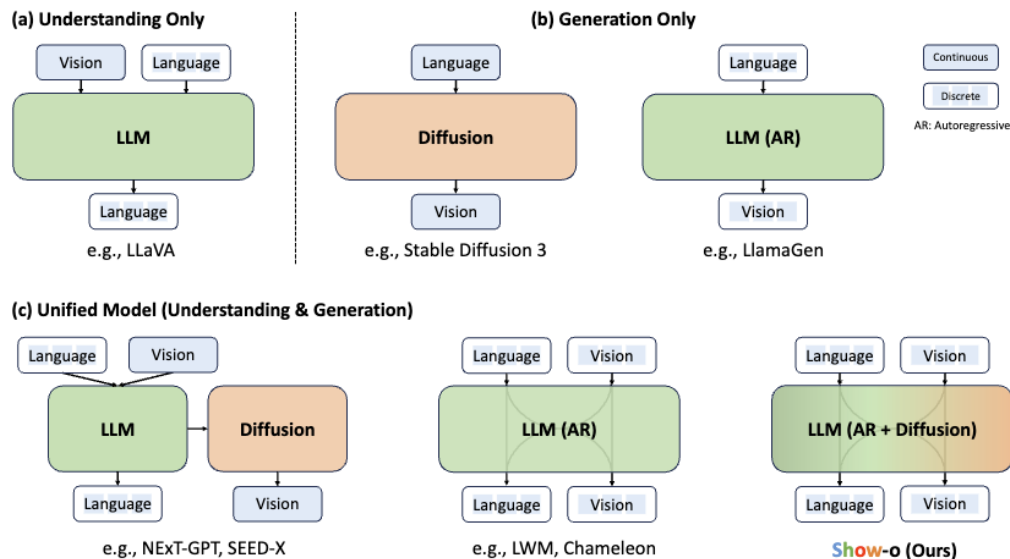
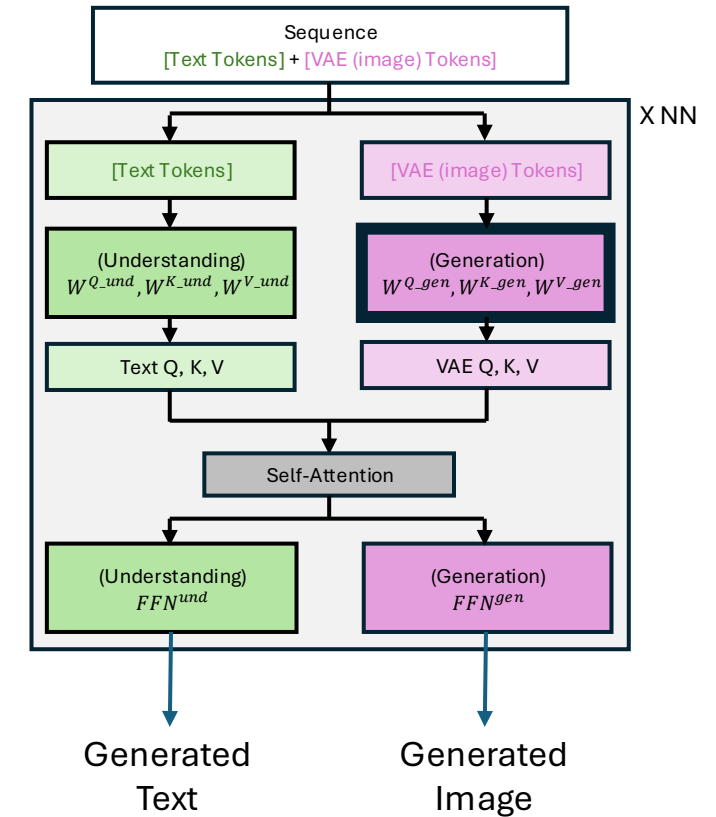


Figure 1: Characteristics comparison among understanding only, generation only, and unified (understanding & generation) models. “Vision” and “Language” indicate the representations from specific input modalities. In this context, “Diffusion” represents both continuous and discrete diffusion.



Base Model Architecture

- **Applicability Assessment Criteria**

- Whether generation-side tokens or latents can be separated
- Whether they meet the text stream in a shared context
- Whether hook-based intervention can be restricted to the generation side

- **Models by Release Order and Applicability Assessment**

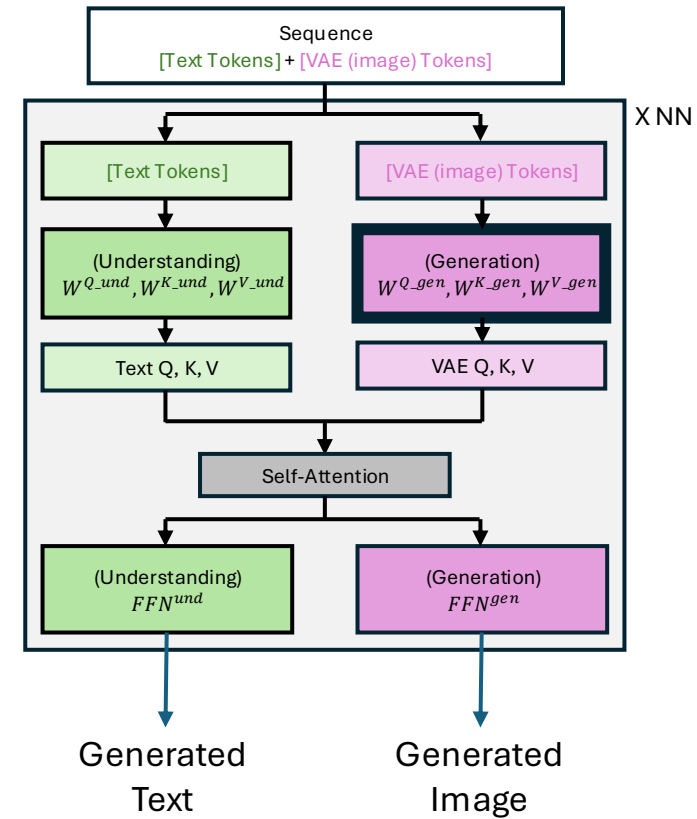
Model	Applicability	Github Released	Title	Citations	Release Date
Transfusion	△	○	Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model	496	2024-08-20
Show-o	△	○	Show-o: One Single Transformer to Unify Multimodal Understanding and Generation	631	2024-08-22
MonoFormer	△	○	MonoFormer: One Transformer for Both Diffusion and Autoregression	50	2024-09-24
JanusFlow	△		JanusFlow: Harmonizing Autoregression and Rectified Flow for Unified Multimodal Understanding and Generation	198	2024-11-12
LMFusion	△		LMFusion: Adapting Pretrained Language Models for Multimodal Generation	27	2024-12-19
Nexus-Gen	X		Nexus-Gen: Unified Image Understanding, Generation, and Editing via Prefilled Autoregression in Shared Embedding Space	4	2025-04-30
Mogao	○		Mogao: An Omni Foundation Model for Interleaved Multi-Modal Generation	71	2025-05-08
BAGEL	○	○	Emerging Properties in Unified Multimodal Pretraining	486	2025-05-20

Base Model Architecture

- Models by Release Order and Applicability Assessment

Model	Applicability	Github Released	논문	Citations	Release Date
OpenUni	X		OpenUni: A Simple Baseline for Unified Multimodal Understanding and Generation	22	2025-05-29
Show-o2	△		Show-o2: Improved Native Unified Multimodal Models	132	2025-06-18
LightFusion	O		LightFusion: A Light-weighted, Double Fusion Framework for Unified Multimodal Understanding and Generation	8	2025-10-27
HBridge	X		HBridge: H-Shape Bridging of Heterogeneous Experts for Unified Multimodal Understanding and Generation	2	2025-11-25
TUNA	△		TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models	13	2025-12-01
EMMA	X	O	EMMA: Efficient Multimodal Understanding, Generation, and Editing with a Unified Architecture	4	2025-12-04
InternVL-U	X		InternVL-U: Democratizing Unified Multimodal Models for Understanding, Reasoning, Generation and Editing	2	2026-03-10

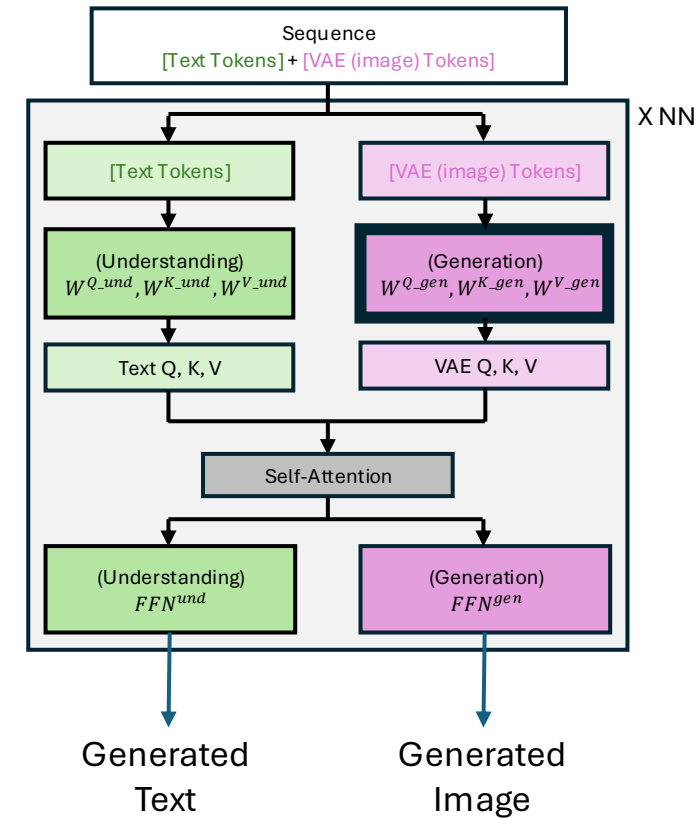
Problem Statement



Problem Statement

- **Scenarios**

- Celebrity or private faces
- Copyrighted characters
- Artist-specific styles
- IDs and private documents
- Graphic violence or sexual concept

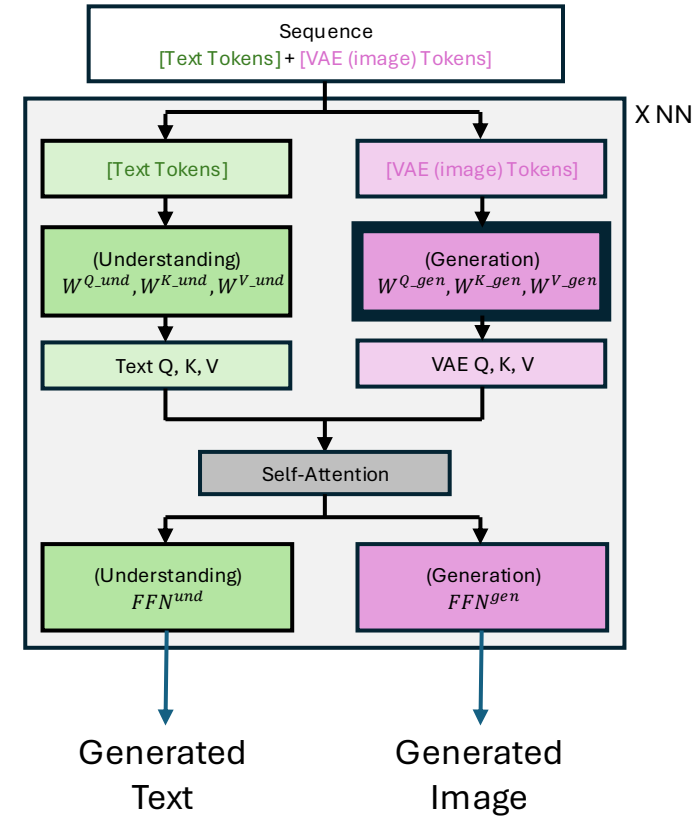


Problem Statement

- **Scenarios**

- Celebrity or private faces
- Copyrighted characters
- Artist-specific styles
- IDs and private documents
- Graphic violence or sexual concept

→ *Knows but can't draw...*



Contributions (Goals)

- The first work unlearning work on UMM
- The first work of modality selective unlearning
- Using SAE to 이미지 손상 없이 generate 하는 방법. (이거 자체는 이미 있던 방법임)
 - Added Counterfactual prompts to 다른 개념과 더 잘 분리하기 위해서.

Method: Contrastive Feature Selection

1. Unsupervised SAE Training

- BAGEL의 VAE-side hidden state에 대해 TopK SAE를 학습
- 이 단계에서는 Concept label을 사용하지 않음

UnlearnCanvas

20 object classes

× class당 80 anchor prompts

× (51 style postfix + 1 base prompt)

= 83,200

2. Matched prompt set 구성

- 같은 scene에 대해 target, background, retain prompt를 만든다.
 - Target: A horse in a prairie.
 - Background: An empty prairie.
 - Retain: A cow in a prairie.
 - Retain: A deer in a prairie.

(Method)

• Unsupervised Learning on SAE

- **Loss:** $L_{\text{unsup}} = \text{FVU} + \alpha_{\text{aux}} * L_{\text{aux}}$
 - $\text{FVU} = ||\hat{x} - x||^2 / ||x - \text{mean}(x)||^2$
 - $\alpha_{\text{aux}} = 0.03125$
 - $L_{\text{aux}} \approx ||\text{decode}(\text{dead-latent activations}) - \text{residual}||^2 / \text{total_variance}$

• SAE Forward:

- $v = W_{\text{enc}}(x - b_{\text{pre}}) + b_{\text{enc}}$
- $z = \text{TopK}(\text{ReLU}(v), k=32)$
- $\hat{x} = W_{\text{dec}} z + b_{\text{pre}}$

Let $\mathbf{x} \in \mathbb{R}^d$ denote the d -dimensional vector of activations from a single position of a feature map and let n be the latent dimension in sparse autoencoder. The encoder and decoder of standard single-layer ReLU sparse autoencoder (Bricken et al., 2023) are then defined as follows:

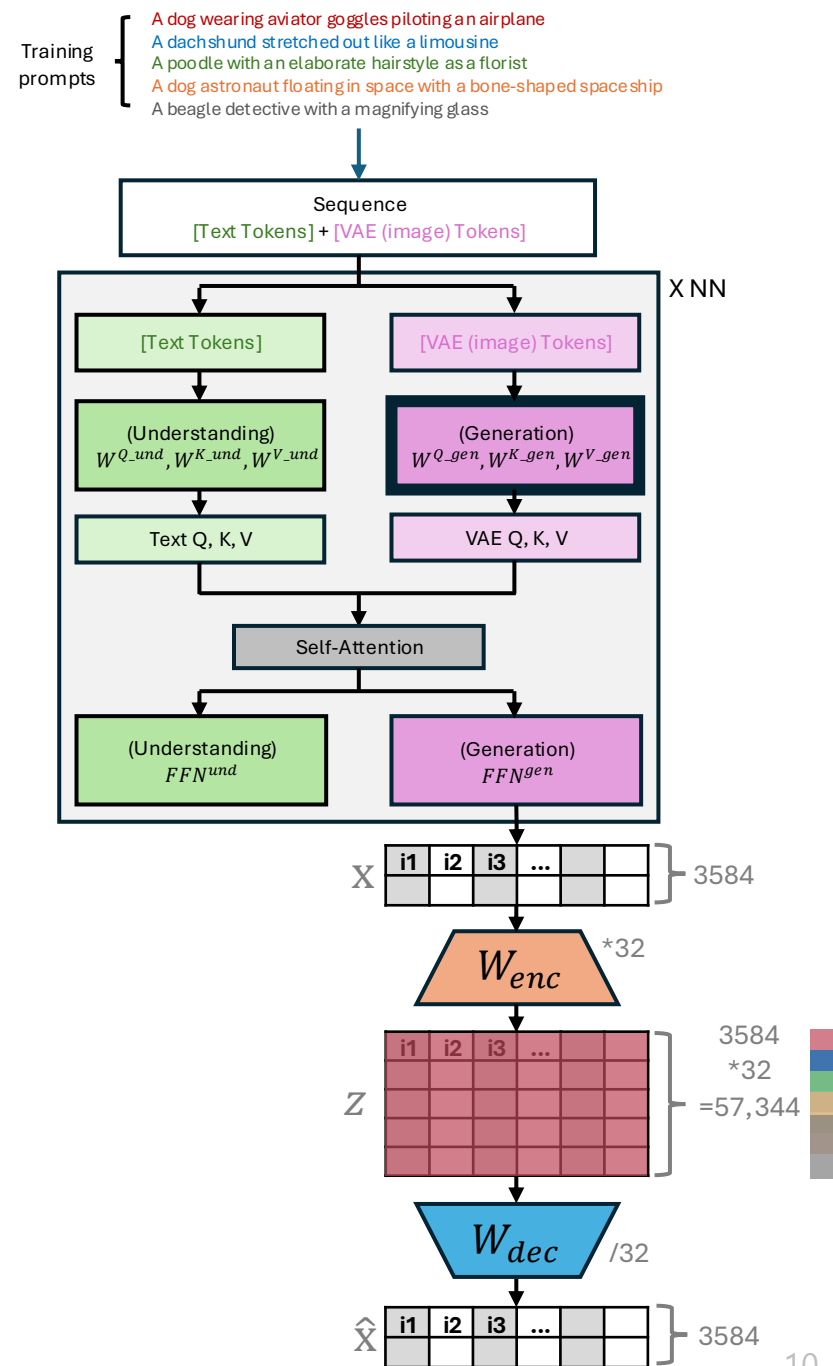
$$\begin{aligned} \mathbf{z} &= \text{ReLU}(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}) \\ \hat{\mathbf{x}} &= W_{\text{dec}} \mathbf{z} + \mathbf{b}_{\text{pre}}, \end{aligned} \quad (1)$$

where $W_{\text{enc}} \in \mathbb{R}^{n \times d}$ and $W_{\text{dec}} \in \mathbb{R}^{d \times n}$ are encoder and decoder weight matrices respectively, $\mathbf{b}_{\text{pre}} \in \mathbb{R}^d$ and $\mathbf{b}_{\text{enc}} \in \mathbb{R}^n$ are learnable bias terms. Elements of $\mathbf{z}$ called feature activations are usually denoted as $f_{1,\dots,n}(\mathbf{x})$. Typically, n is equal to d multiplied by a positive *expansion factor*.

The objective function of SAE is defined as:

$$\mathcal{L}(\mathbf{x}) = ||\mathbf{x} - \hat{\mathbf{x}}||_2^2 + \alpha \mathcal{L}_{\text{aux}}, \quad (2)$$

where $||\mathbf{x} - \hat{\mathbf{x}}||_2^2$ is a reconstruction error and $\mathcal{L}_{\text{aux}}$ is a reconstruction error using only the largest k_{aux} feature activations that have not fired on a large number of training samples, so-called *dead latents*. The auxiliary loss is used to prevent dead latents from occurring and is scaled by a coefficient α .



Method: Contrastive Feature Selection

3. Feature activation 계산

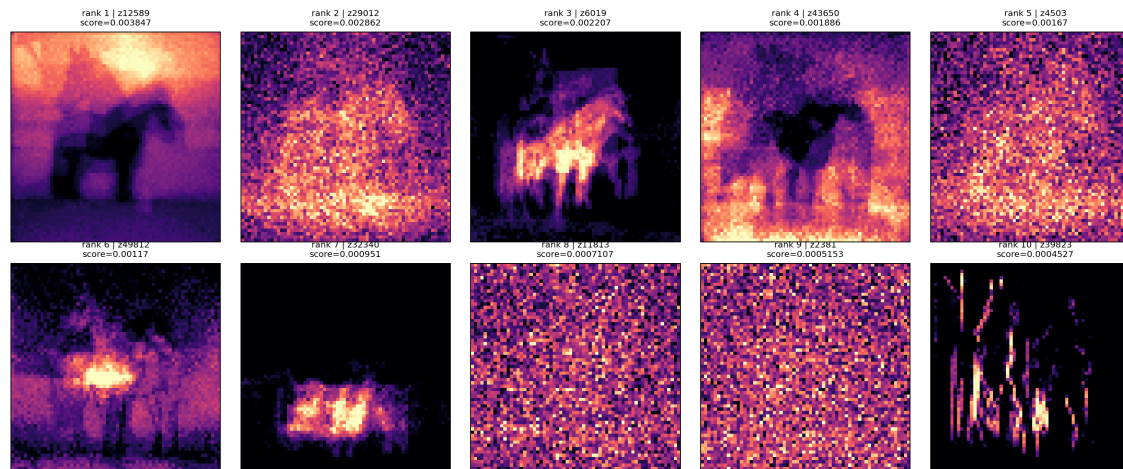
- 각 SAE feature z_i 에 대해 target, background, retain에서의 activation을 계산
 - T_i = activation on target prompts
 - B_i = activation on background prompts
 - R_i = activation on retain prompts

4. Contrastive score 계산

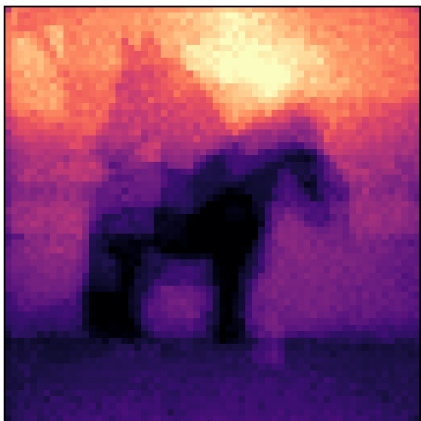
- feature score를 다음처럼 정의: $\text{score}_i = T_i - \lambda B_i - \mu R_i$ * preservation
- 즉 horse에서는 강하지만, empty/cow/deer에서는 약한 feature가 높은 점수를 받는다.

5. Feature bundle 선택

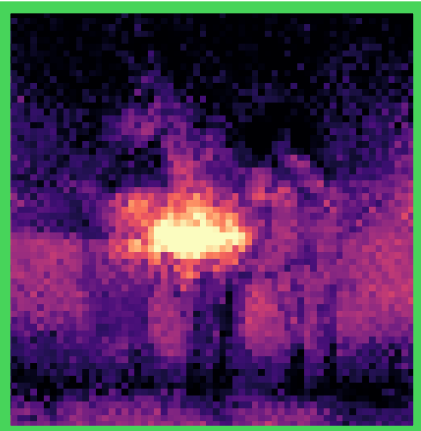
- score가 높은 feature를 작은 bundle로 선택한다.
- 이때 서로 너무 비슷한 feature는 줄이고, target coverage가 늘어나는 feature를 우선한다.



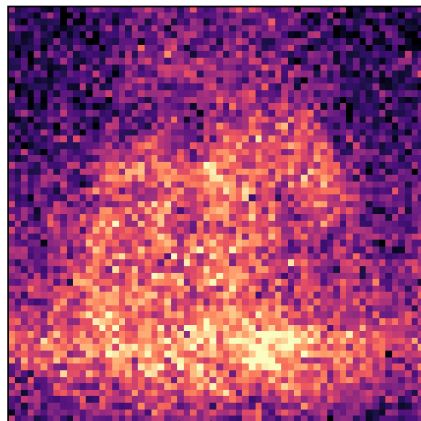
rank 1 | z12589
score=0.003847



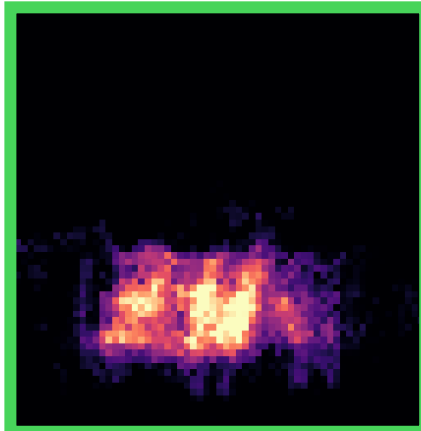
rank 6 | z49812
score=0.00117



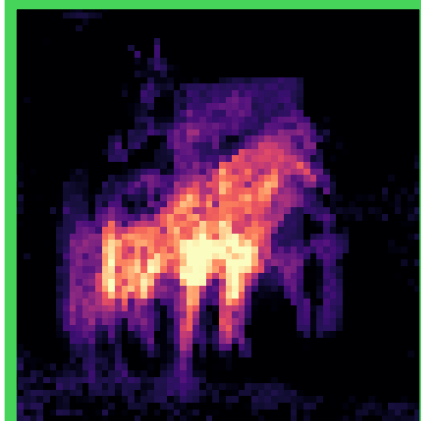
rank 2 | z29012
score=0.002862



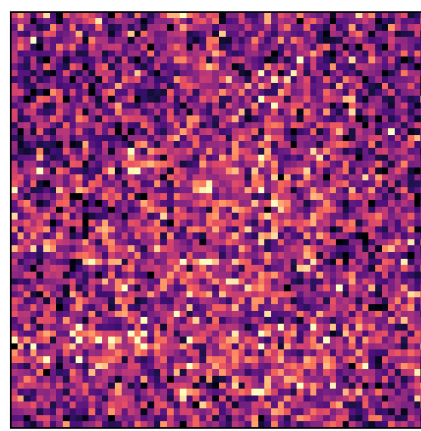
rank 7 | z32340
score=0.000951



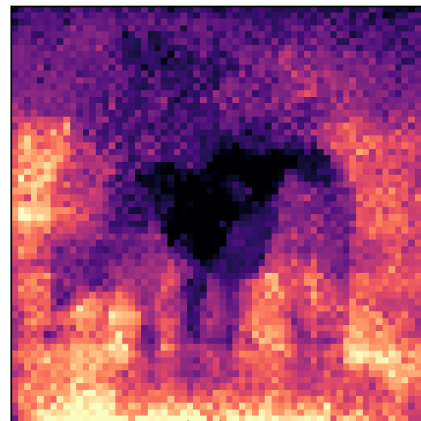
rank 3 | z6019
score=0.002207



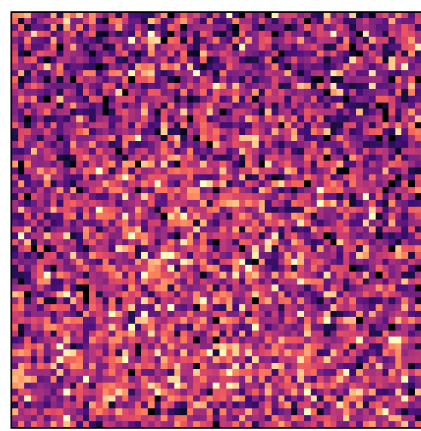
score=0.0007107



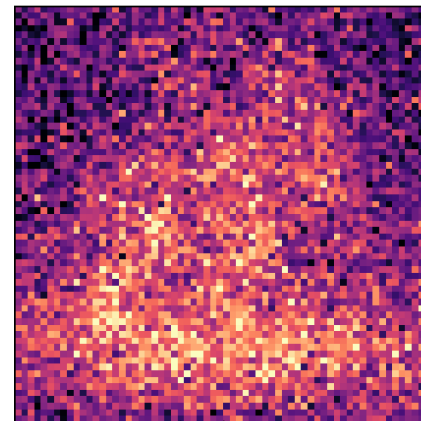
rank 4 | z43650
score=0.001886



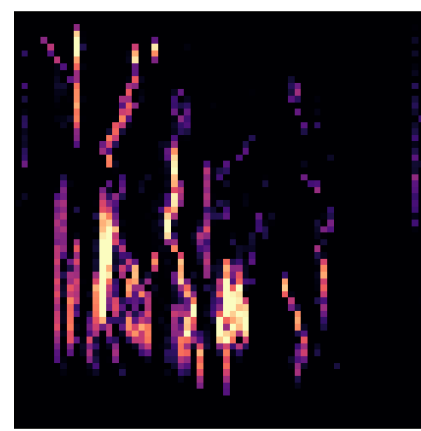
rank 9 | z2381
score=0.0005153



rank 5 | z4503
score=0.00167



rank 10 | z39823
score=0.0004527



Method: Contrastive Feature Selection

6. Threshold와 suppression strength 설정

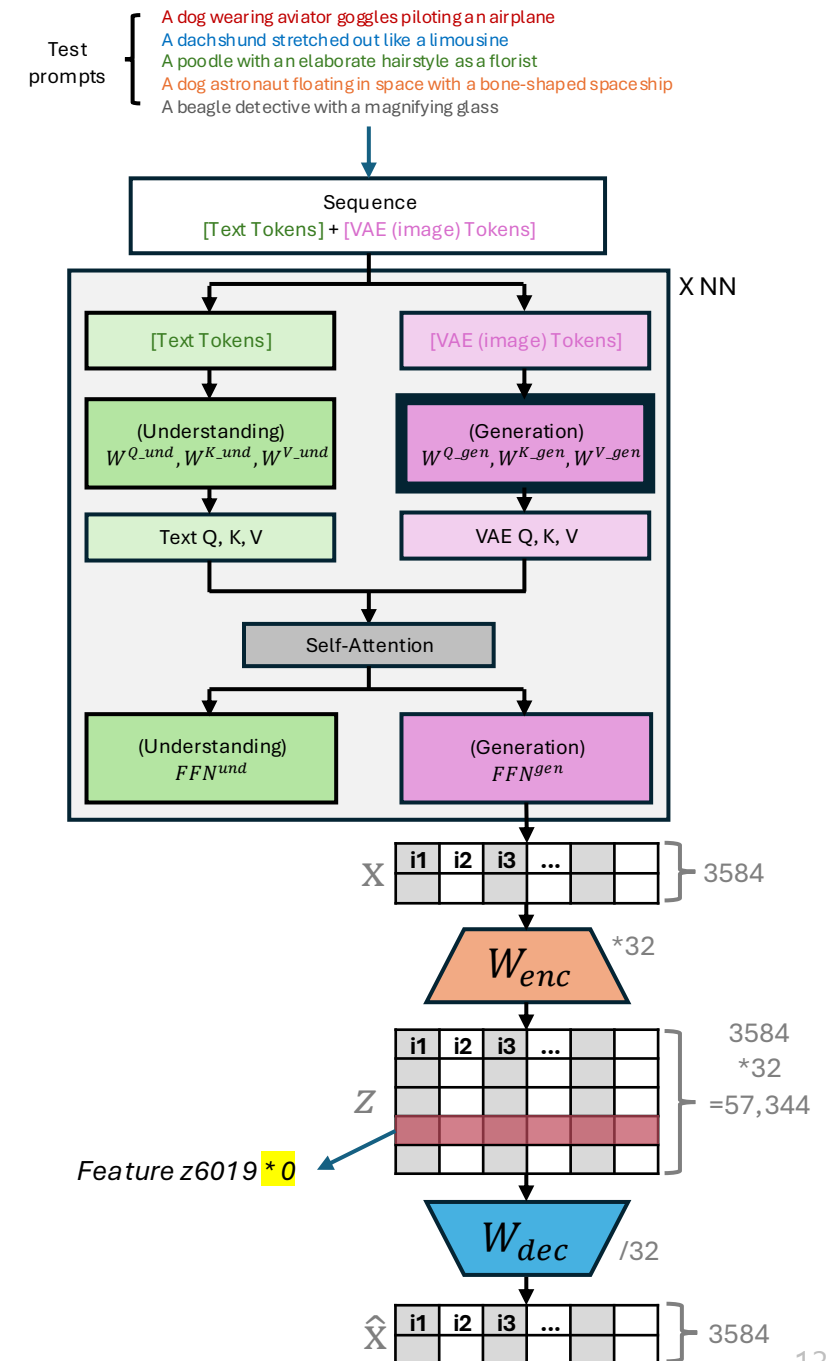
- retain/background activation 분포를 기준으로 threshold를 정한다.
- 이후 validation prompt에서 suppression strength gamma를 조절한다.

7. Concept feature bank 저장

- 최종적으로 concept별로 다음 정보를 저장한다.
 - concept: Horses
 - features: [z6019, ...]
 - threshold
 - gamma
 - layer
 - denoising steps

8. Inference-time intervention

- 이미지 생성 중 선택된 feature가 threshold보다 강하게 켜질 때만 suppress한다.

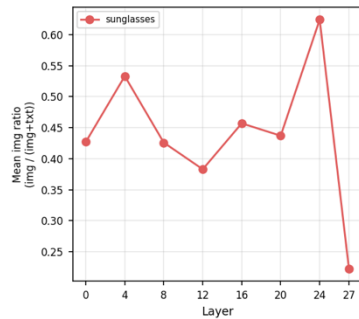


Methods (In Short)

1. Collect prompts for the target concept.
2. Run BAGEL and cache VAE-side hidden states.
3. Train an unsupervised TopK SAE on the cached states.
4. Rank SAE features by target-concept activation.
5. Use counterfactual prompts to filter concept-specific features.
6. Select candidate features, such as z6019 for Horses.
7. Suppress selected features during image generation.
8. Evaluate target suppression and image preservation.

Method (Additional Consideration)

- But which layer to add SAE?
 - 24 layer
 - Cross Attention Map 그려보니까 semantic하게 두 토큰이 연결되어있는 부분이 24 레이어여서, 거기에 SAE를 달아 sae가 semantic meaning을 잘 잡도록 의도했다.

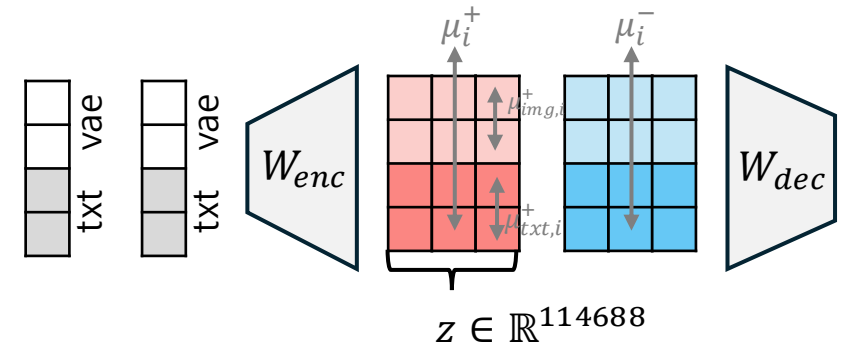


$$ImgRatio_i = \frac{\mu_{img,i}^+}{\mu_{img,i}^+ + \mu_{txt,i}^+ + \epsilon}$$

- 1에 가까울수록 해당 feature가 img 토큰에서만 활성화
→ 안전하게 억제 가능 (높을수록 좋음)
- 0.5에 가까우면 shared feature
→ 억제 시 txt도 영향받음

$$\mu_{img,i}^+ = \frac{1}{N_{img}} \sum_{img_tokens} Z_i$$

$$\mu_{txt,i}^+ = \frac{1}{N_{txt}} \sum_{txt_tokens} Z_i$$



- And How much it cost?
 - Time

Remained Research Questions / Discussions so far

- Not just text prompt (+ think mode)
 - 이렇게 하면 모델이 생각하는 개념의 특징을 알아낼 수 있음.
 - 근데 또 모델이 그거 이외의 특징을 스스로 생각해서 만들어낼 수도 있음.
- Not clearly 구별되는 객체와 배경 when inference.
- What if the object is at the corner (not general cases)
- 어쨌든 완전한 unlearning이 아니라 blocking / filtering에 가까움.

- 찾은 feature들의 조합을 x0을 해서 intervention 후 생성 결과를 보고싶었음.
- 그 결과에서 말이 조금이라도 지워진다면 성공 가능성이 있는 거니까.
- 그러나 연세대 서버에서 queue를 걸어놓고 gpu를 잡는 시스템인데
- 요즘 사람들이 많이 사용하는지 이틀을 기다려도 gpu가 안잡혀서 이미지 결과를 생성해보지는 못했다.

그거 말고 관심있는 분야

- Generative Models
- Computer Vision
- Multimodal Models
- Interpretability

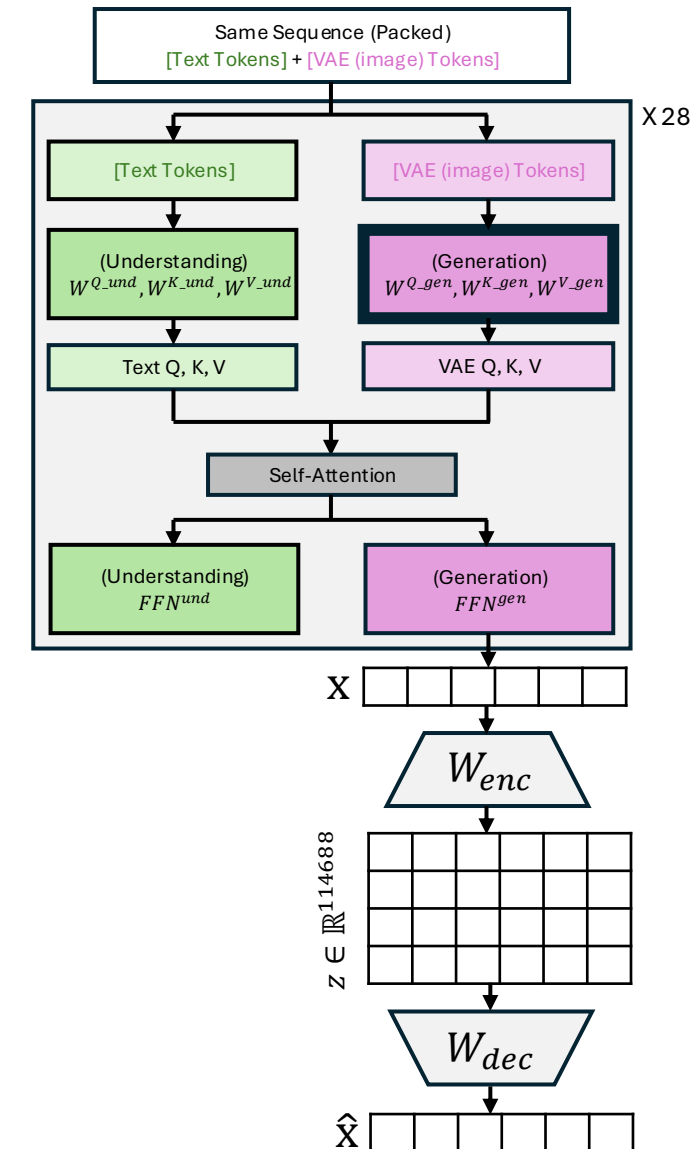


Recap: Loss

- VAE-only SAE
- $L = L_{unsup} + \beta L_{bind} + \eta L_{route}$
 1. L_{unsup} 로 SAE를 먼저 학습한다.
 2. Route-aware score matrix $S(c, i)$ 를 계산한다.
 3. $S(c, i)$ 를 사용해 concept-to-latent assignment i_c 또는 bundle G_c 를 만든다.
 4. 그 assignment를 고정한 뒤 L_{bind} , L_{route} , L_{neg} 로 supervised shaping을 수행한다.

- $L_{unsup} = ||x - \hat{x}||_2^2 + \alpha L_{aux}$
 - $||x - \hat{x}||_2^2$: reconstruction error
 - L_{aux} : reconstruction error using only the largest k_{aux} feature activations that have not fired on a large number of training samples, so-called dead latents.

$$L = L_{unsup} + \beta L_{bind} + \eta L_{route}$$

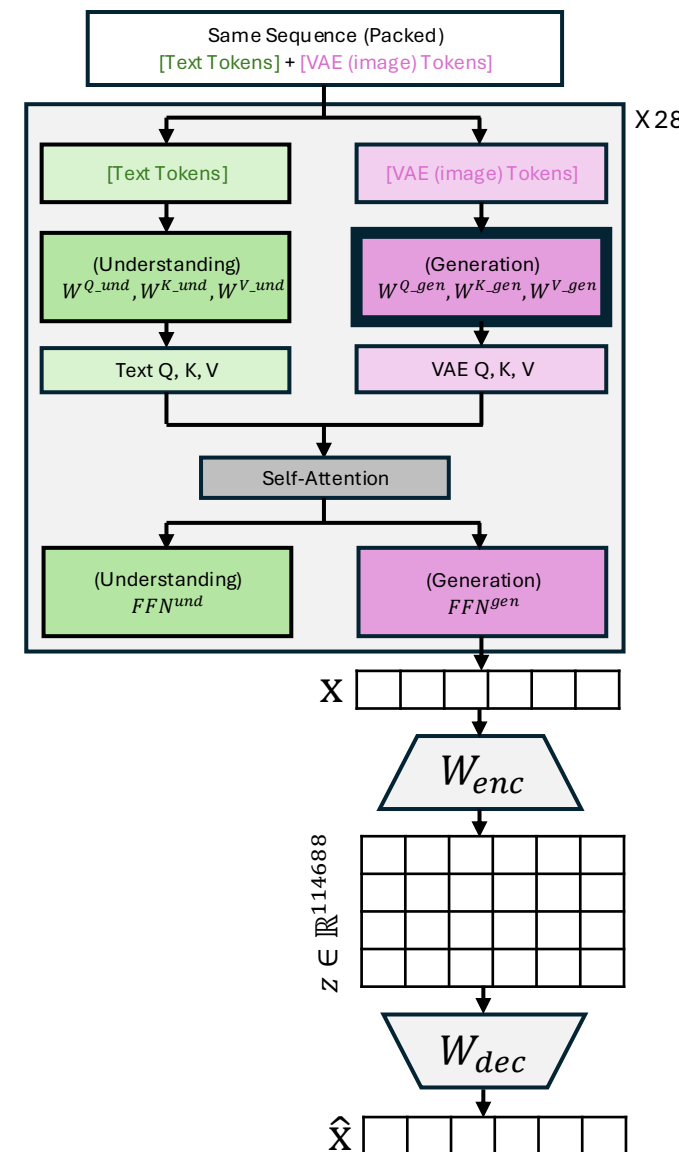


Recap: Loss

- Concept c 와 latent i 의 적합도: $S(c, i) = \frac{\bar{\mu}_{c,i}^+}{\sum_{j=1}^n \bar{\mu}_{c,j}^+ + \delta} - \frac{\bar{\mu}_{c,i}^-}{\sum_{j=1}^n \bar{\mu}_{c,j}^- + \delta}$
 - $\bar{\mu}_{c,i}^+$: concept c 가 present인 sample들에서 latent i 의 평균 반응
 - $\bar{\mu}_{c,i}^-$: concept c 가 absent인 sample들에서 latent i 의 평균 반응
 - n : SAE latent의 총 개수
 - δ : 분모나 0에 가까워지는 것을 막는 작은 상수
- 이 score를 기반으로 각 concept에 대한 하나의 latent를 할당함
 - positive에서 강하고
 - negative에서 약하고
 - 같은 concept 내부에서 상대적으로 비중이 크고
 - negative sample 쪽에서는 상대적으로 비중이 작은 latent

- $L_{bind} = -\frac{1}{|R|} \sum_{r \in R} \frac{1}{|\mathcal{B}_r|} \sum_{b \in \mathcal{B}_r} \frac{1}{|C_b^{(r)}|} \sum_{c \in C_b^{(r)}} \log \sigma(p_V^{(r,b)}(v)_{i_c})$
 - $R = \{text \rightarrow vae, text + img \rightarrow vae\}$: route 집합
 - $\mathcal{B}_r$: route r 에서의 mini-batch
 - $b \in \mathcal{B}_r$: route r 에서의 batch 안 sample
 - $C_b^{(r)}$: sample b 에 실제로 포함된 positive concept 집합
 - i_c : concept c 에 할당된 latent index
 - $p_V^{(r,b)}(v)$: route r , sample b 에서의 VAE-side pooled preactivation

$$L = L_{unsup} + \beta L_{bind} + \eta L_{route}$$



Recap: Loss

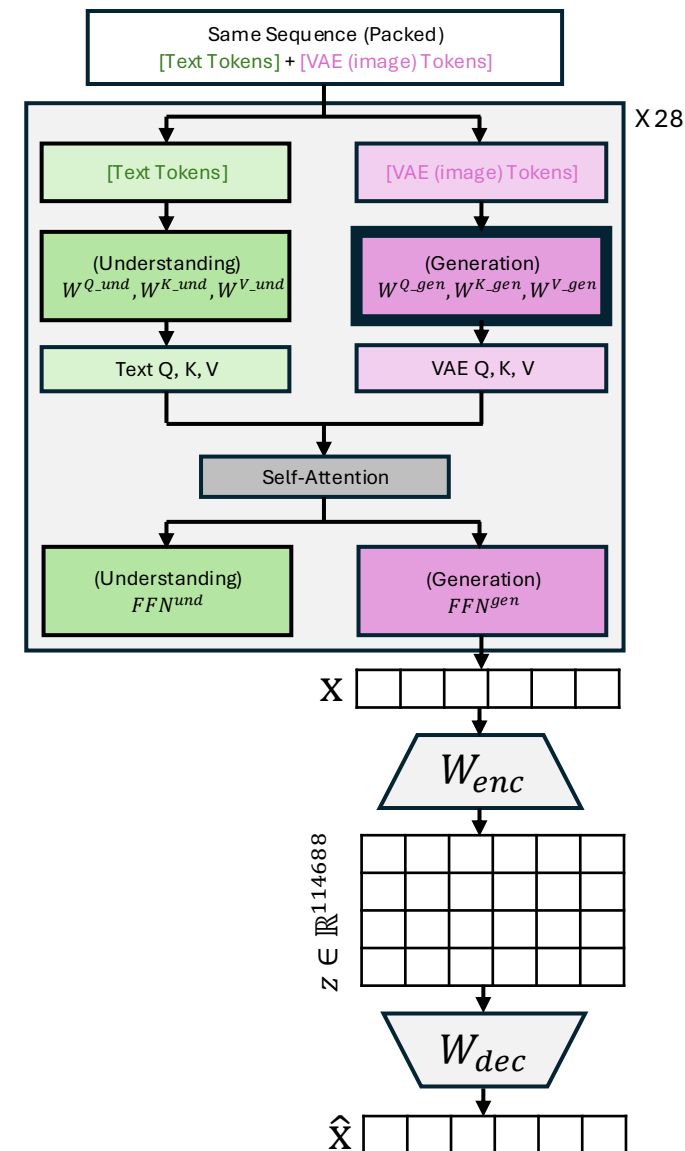
- $L_{route} = \frac{1}{|c|} \sum_c \left(\mu_c^{(text \rightarrow vae)} - \mu_c^{(text+img \rightarrow vae)} \right)^2$
 - C : 전체 concept 집합
 - $\mu_c^{(r)}$: route r 에서 concept c 의 assigned latent 평균 반응
 - 의미
 - 같은 concept c 가
 - Text $\rightarrow$ vae
 - Text + img $\rightarrow$ vae 두 경로 중 어디로 들어오더라도
 - Assigned latent i_c 의 VAE-side 반응이 비슷하도록 만든다

→ 둘 다 낮아질 수 있는 문제?

$$\mu_c^* = \text{stopgrad} \left(\max \left[\mu_c^{(text \rightarrow vae)}, \mu_c^{(text+img \rightarrow vae)} \right] \right)$$

$$L_{route}^{up} = \frac{1}{|c|} \sum_{c \in C} \left[\text{ReLU} \left(\mu_c^* - \mu_c^{(text \rightarrow vae)} \right)^2 + \text{ReLU} \left(\mu_c^* - \mu_c^{(text+img \rightarrow vae)} \right)^2 \right]^2$$

$$L = L_{unsup} + \beta L_{bind} + \eta L_{route}$$



Experiment (0) Settings

- UnlearnCanvas

- 20 object classes

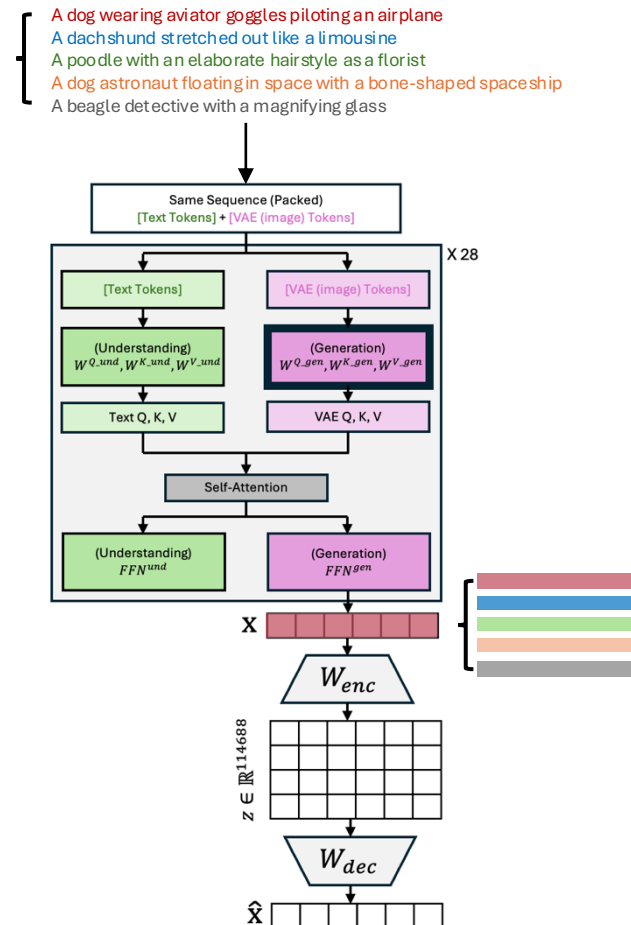
× class당 80 anchor prompts

× (51 style postfix + 1 base prompt)

= 83,200

[Collecting]

Took 10 hours to collect 1600 object class prompts



[Train]

Less than 2h
(10 epochs)

$$L$$

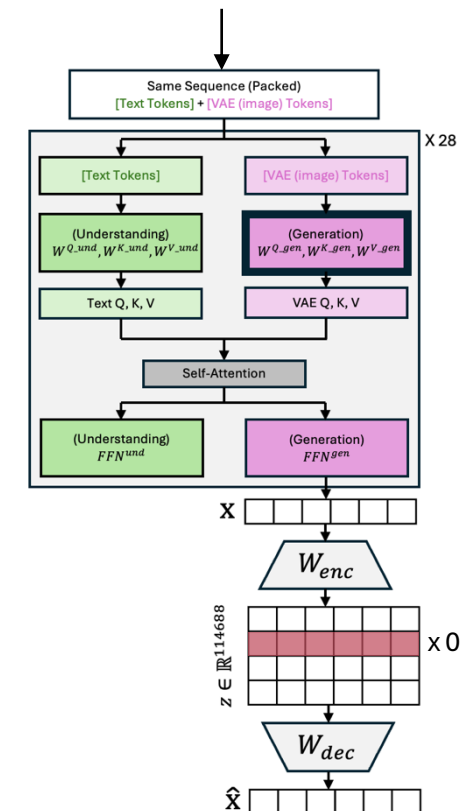
$$= L_{\text{unsup}}$$

$$+ \beta L_{\text{bind}}$$

$$+ \eta L_{\text{route}}$$

[Test]

An image of Dogs in {theme 1} style.
 An image of Dogs in {theme 2} style.
 An image of Dogs in {theme 3} style.
 An image of Dogs in {theme 4} style.
 An image of Dogs in {theme 5} style.



Extension to Image Editing

- T2I
 - UnlearnCanvas
 - I2P
 - EMMA
 - HUB
- I2I
 - MagicBrush
 - I2EBench
 - Pico-banana-400k